

Ahmed Nasser Eldin

AI Engineer — Garment Manufacturing & Computer Vision

Cairo, Egypt · Open to on-site full-time roles in Cairo

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AN

AI Engineer with 2+ years of practical experience building Computer Vision, machine learning, and cloud solutions—combined with seven years of hands-on garment production experience operating Singer, overlock, and coverstitch machines. I understand both the technical AI pipeline and the daily realities of garment lines, allowing me to design practical systems for production visibility, workforce analytics, quality, and material movement.

7 years

GARMENT PRODUCTION FLOOR

2+ years

APPLIED AI EXPERIENCE

10+ projects

AI AND CLOUD SYSTEMS

INDUSTRIAL AI PROJECTS

FactoryVision — Workforce Presence & Active-Time Intelligence

2026

Computer Vision system demonstrator

- Designed an edge-first concept for turning garment-factory camera feeds into attendance, presence, and active-time events.
- Included tracking, privacy controls, confidence thresholds, exception review, and shift-reporting workflows.
- Built an interactive portfolio demonstrator with synthetic workers and events; proposed stack includes Python, OpenCV, YOLO, FastAPI, PostgreSQL, and Edge AI.

StitchOps — Garment Workforce & Production Analytics

2026

Interactive analytics dashboard

- Developed an interactive dashboard connecting paid labor hours to garment output, efficiency, overtime, target attainment, and defect rates.
- Implemented department and period filters, dynamic KPI cards, visual comparisons, operational insights, and CSV export.
- Used deterministic synthetic data across sewing, cutting, finishing, and packing departments.

Bulbul — Mobile Material-Flow Assistant

Prototype

Physical delivery robot + garment application case study

- Built a physical wheeled robot prototype originally intended to deliver medicine while reducing contact around infectious-disease patients.
- Adapted the mobility concept for moving cut panels and garment bundles between production stages.
- Created a transparent case study separating the documented prototype from the proposed factory application.

CORE EXPERTISE

Computer Vision Object Detection
Tracking Image Processing OCR
Edge AI Machine Learning
Deep Learning Production Analytics
AI Deployment

TECHNICAL SKILLS

Languages

Python, SQL, C++, Java

AI & Vision

PyTorch, TensorFlow, OpenCV, YOLO, Scikit-learn, Tesseract OCR, Hugging Face

Engineering

FastAPI, Docker, Kubernetes, Git, Linux, CI/CD, REST APIs

Cloud & Edge

AWS, Azure, Raspberry Pi, Mobile AI Deployment

Data

PostgreSQL, MySQL, MongoDB, Power BI, Streamlit

RELEVANT EXPERIENCE

Garment Sewing Machine Operator

2019 — Jul 2026

Singer · Overlock · Coverstitch

- Seven years of hands-on garment production experience operating industrial sewing machines.
- Developed practical understanding of production rhythm, line handoffs, operator effort, quality pressure, and shop-floor constraints.

AI & Computer Vision Intern · Dark Pyramid

Feb — Jun 2026

- Trained classification models across 1,000+ classes and developed small-object detection pipelines for fine-grained symbols.
- Applied annotation, preprocessing, evaluation, PyTorch, TensorFlow, OpenCV, and YOLO.

AI Instructor · Resala Training Centers

Jan — Mar 2025

- Taught Python and Machine Learning through hands-on projects, supporting 12 students.

ADDITIONAL SELECTED PROJECTS

Pathira — Real-Time AI Vision Assistant

Camera streaming, object detection, mobile/edge deployment, voice feedback, Raspberry Pi, TFLite, and React Native.

Face Recognition Attendance & Automation Suite

Attendance tracking with Python and OpenCV, integrated with UiPath automation.

OCR Enhancement Pipeline

Denoising, binarization, contrast adjustment, deskewing, and Tesseract OCR for IDs and forms.

EDUCATION

Bachelor of Artificial Intelligence

Helwan International Technological University

Sep 2022 — Jul 2026

GPA: 3.5

TRAINING & CERTIFICATION

Digital Egypt Pioneers Initiative

Data Science Trainee · Oct 2024 — May 2025

Huawei HCCDA-AI

AI theory, application development, and Huawei Cloud EI services.

Responsible AI approach: workforce monitoring should support operational review, include clear worker policies, and never turn low-confidence model output directly into penalties.